

MOHSEN MOSTAFA

Independent Researcher — Physical & Geometric AI | AI Team Lead & Research Engineer

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RESEARCH STATEMENT

I define and develop **Physical & Geometric AI** — a unified methodology for building robust, trustworthy deep learning systems under computational constraints. My research integrates three pillars: (1) **geometric foundations** (proving how networks adapt to data manifolds), (2) **uncertainty-aware architectures** (Bayesian normalization and activation functions), and (3) **physics-informed optimization** (curvature-aware optimizers for scientific ML). I completed six first-author papers forming this methodology in under 10 weeks using only free cloud GPUs (Kaggle/Colab: NVIDIA T4 15GB, P100 16GB), demonstrating that high-impact research is achievable without massive computational resources.

EDUCATION

B.Sc. Management Information Systems 2001 – 2005

High Institute for Specialized Technological Studies, Cairo, Egypt

RESEARCH EXPERIENCE

Independent Researcher — Physical & Geometric AI | Self-directed 2023 – Present

Founded and executed a coherent research program resulting in six first-author publications within 10 weeks, all validated on free cloud GPUs (Kaggle/Colab).

- ▶ Defined the "Physical & Geometric AI" methodology, unifying geometric deep learning theory, uncertainty-aware architectural components, and physics-informed optimization.
- ▶ Proved four fundamental theorems on how gradient descent shapes neural representations on data manifolds.
- ▶ Developed novel Bayesian normalization layers (R-LayerNorm, Bayesian R-LayerNorm) and activation functions (Bayesian PASA) with built-in uncertainty quantification.
- ▶ Created EPANG-Gen, a memory-efficient curvature-aware optimizer that eliminates catastrophic training failures on stiff PDE problems.
- ▶ Validated all methods on challenging benchmarks: turbulent flow, CIFAR-10-C, and Darcy flow — under strict computational constraints.

PUBLICATIONS & PREPRINTS

Geometric Foundations

Mostafa, M. — *Geometric Foundations of Racing Dynamics: How Gradient Descent Adapts Network Capacity on Data Manifolds with Application to Bayesian R-LayerNorm.* (2026). Q1 Journal (Preprint).

Mostafa, M. — *Gaussian-Constrained LeJEPa Representations for Unsupervised Scene Discovery in the Image Matching Challenge 2025*. (2025). arXiv:2602.07016. [\[arXiv\]](#)

Mostafa, M. — *HPST: A Hybrid Physics-Spectral-Threshold Framework for Fluid Flow Analysis with Theorem Proving and Graph Neural Networks*. (2026). Preprint. [\[Preprint\]](#)

Uncertainty-Aware Architectures

Mostafa, M. — *Bayesian R-LayerNorm: Uncertainty-Aware Adaptive Normalization with Provable Robustness Bounds*. (2026). Q1 Journal (Preprint).

Mostafa, M. — *Bayesian PASA: Probabilistic Adaptive Sigmoidal Activation with Uncertainty Quantification*. (2026). Q1 Journal (Preprint).

Mostafa, M. — *Uncertainty-Aware Classifier with Physics-Based Rejection (UA-PBR): A Proof-of-Concept under Computational Constraints*. (2026). Q1 Journal (Preprint).

Mostafa, M. — *R-LayerNorm: Robust Layer Normalization with Adaptive Noise Suppression for Noisy Image Data*. (2025). Preprint. [\[Preprint\]](#)

Physics-Aware Optimization

Mostafa, M. — *EPANG-Gen: A Curvature-Aware Optimizer with Uncertainty Quantification for Scientific Machine Learning — A Proof-of-Concept under Computational Constraints*. (2026). Q1 Journal (Preprint).

Under Review

Mostafa, M. — *Geometric Deep Learning for Point Cloud Segmentation*. Under revision for WACV 2027. [\[Under Review\]](#)

Mostafa, M. — *Physics-Informed Neural Networks for Fluid Flow*. (2025). arXiv. [\[arXiv\]](#)

PROFESSIONAL EXPERIENCE

AI Team Lead & Computer Vision Researcher | [GraviLog \(Remote — US/UAE\)](#) Sep 2025 – Present

- ▶ Designed end-to-end multimodal AI system architecture (CV + LLM) with TensorRT/ONNX optimizations achieving 40% latency reduction.
- ▶ Defined technical roadmap for 3 products and delivered strategic presentations to C-level stakeholders.
- ▶ Led RAG retrieval pipeline (Pinecone + embeddings); mentored 4 engineers in MLOps and system design.

Senior AI Developer (Research Focus) | [OMC Technologies \(Remote — Limerick, Ireland\)](#) May 2023 – Aug 2025

- ▶ Built AI agents integrating LLMs (Falcon, DeepSeek, Qwen, Llama) — automated 3 workflows, reducing manual effort by 40%.
- ▶ Deployed production RAG system (LangChain + Pinecone) reducing hallucinations by 35%.
- ▶ Multimodal healthcare AI system: F1 score +19%; latency reduced from 210ms to 145ms via streaming + FP16.
- ▶ Fraud detection system: AUC +17%, false positives –22%; FastAPI service with p99 ≤ 95ms at 500 RPS.
- ▶ Managed AI infrastructure with 24 GPUs across 12 nodes for real-time IoT anomaly detection.
- ▶ Mentored 5 junior engineers, leading to 2 promotions.

AI Developer & Researcher | [Nexgro \(South Africa\)](#) Apr 2020 – Apr 2023

- ▶ Grain quality CV pipeline achieving 98% accuracy across 6 grain types.
- ▶ Drone/satellite anomaly detection (NDVI + YOLOv8): manual inspection efficiency +160%.
- ▶ Optimized 3D point cloud segmentation for LiDAR/DSM data achieving sub-300ms inference on edge devices.
- ▶ Implemented TensorRT/ONNX optimizations delivering 3× speedup in model inference.
- ▶ LLM RAG assistant on technical manuals — saved 15 person-hours/week.

Data Scientist | Nexgro (South Africa) Mar 2018 – Mar 2020

- ▶ Predictive ML models for land use classification, crop yield estimation, and water consumption forecasting.
- ▶ Forecast models reduced logistics costs by 18%; predictive maintenance prevented 3 breakdowns (~\$45K saved).
- ▶ Engineered ETL pipelines reducing data processing time by 20%.

Systems Administrator | Egyptian Co. Jun 2013 – Mar 2018

- ▶ IT infrastructure management for 200+ users → 99.8% uptime; multi-site network & servers.
- ▶ Automation scripts reduced manual ticket handling by 25%.
- ▶ Self-directed pivot to data science (certifications 2016–2018); managed LAN/WAN, firewalls, backup for 150+ nodes.

IT Network Administrator | International Co. for Trade Aug 2009 – Jun 2013

- ▶ Administered LAN/WAN, firewalls, and backup solutions for 150+ nodes.

IT Network Specialist | International Co. for Trade Aug 2005 – Jun 2009

- ▶ First-line network support, hardware procurement, and vendor coordination.

TECHNICAL SKILLS

Deep Learning	PyTorch, TensorFlow, Hugging Face (Falcon, Llama, Mistral, Qwen), JEPA, CLIP, YOLO, Mask2Former
Probabilistic ML	Bayesian Deep Learning, Uncertainty Quantification, Natural Gradient Methods, Curvature-Aware Optimization
GenAI / LLM	RAG, ColBERT, LangChain, DeepSeek, RLHF concepts, Streaming Inference
Computer Vision	OpenCV, 3D Point Cloud Segmentation (LiDAR/DSM), Multi-View Geometry, Image Processing
MLOps & DevOps	MLflow, Kubeflow, Airflow, Docker, Kubernetes, KServe, FastAPI, GitLab CI, GCP Vertex AI, AWS SageMaker
Streaming / Big Data	Apache Kafka, Spark Structured Streaming, Hadoop HDFS, Parquet
Optimization	TensorRT, ONNX, Model Pruning/Quantization, FP16 Inference
Data & Databases	Python (Pandas, Dask, Polars, PySpark), SQL (PostgreSQL), MongoDB, Redis, Pinecone
Responsible AI	GDPR, AI Ethics, Fairness, Transparency, Uncertainty-Aware Systems

SELECTED CERTIFICATIONS

- ▶ Deep Learning Specialization — DeepLearning.AI
- ▶ TensorFlow Developer & Advanced TensorFlow — Google
- ▶ Google Cloud MLOps — Google
- ▶ Total Data Quality — University of Michigan
- ▶ Research Methodology — University of London
- ▶ IBM Data Science Professional Certificate — IBM
- ▶ CCNA 200-301 & ENCORE 350-401 — Cisco

LEADERSHIP & COMMUNITY

- ▶ Mentored 5+ engineers across OMC Technologies and GraviLog; 2 direct promotions.
- ▶ Conducted internal LLM/MLOps workshops and knowledge-transfer sessions.
- ▶ Designed full MLOps showcase pipeline: Kafka → Spark Streaming → HDFS → MongoDB → PostgreSQL → MLflow → Kubeflow → FastAPI for real-time predictive maintenance.
- ▶ Advocate for resource-constrained AI research; demonstrated 6-paper research program achievable on free-tier cloud GPUs.

LANGUAGES

- ▶ Arabic — Native
- ▶ English — Fluent (C1/C2)